

Single Axis Solar Tracking System

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Abstract: This project presents the design and implementation of a single-axis solar tracking system aimed at enhancing the efficiency of solar panels by ensuring optimal alignment with the sun throughout the day. The system uses an Arduino Uno microcontroller, Light Dependent Resistors (LDRs), and a servo motor to detect the sun's position and adjust the panel accordingly. The primary goal is to demonstrate how solar tracking can improve solar panel exposure without directly generating or storing electricity. The performance of the system is observed under varying weather conditions, and the tracking response is analyzed on an hourly basis for both sunny and cloudy days. This setup provides a cost-effective and practical solution for maximizing solar panel alignment, particularly beneficial for educational and prototype applications.

Keywords: solar panel; tracking; Arduino Uno; LDR; servo motor;

1. INTRODUCTION

In this day, renewable resource is one of the major concerns because of increasing more power demand but the quality and availability of conventional energy sources are not enough. Energy is essential factor for the development of any nations of all over the world. Most of the energy production depends on fossil fuel. The resources of the fossil fuels are limited, so that there are growing demand for energy from renewable resources like solar, geothermal and ocean tidal wave. Among all renewable systems, photovoltaic system is the one which has great chance to replace the conventional energy resources. To enhance the performance of solar panel the only way is to increase the intensity of light falling on it. Solar tracker is the best technology to increase the efficiency of solar panel by keeping panel aligned with the sun's position. The advantage of using solar power for small power generation is its probability; it can be carried whenever or wherever small power generation is required.

In this research, a microcontroller based simple and easily programmed automatic solar tracker is presented. Design and construction of a prototype for solar tracking system detected the sunlight using Light dependent Resistors (LDR), is discussed in this work. The control circuit for the solar tracking system is based on Arduino Uno. This is programmed to detect the sunlight through the LDRs and then actuate the stepper motor to position the solar panel where it receives maximum sunlight. Compared with any other type of motor, the stepper motor is more controllable, more energy efficient, more steady and having high tracking accuracy and suffering little environmental effect.

In this paper, there are mainly five sections to implement this control system. Firstly the system is introduced. The second section describes the system description of overall control system, and some detailed features of the Solar Tracking System. The third and fourth sections present the software, test and result of the overall system. Conclusion is mentioned in the fifth section.

2. SYSTEM DESCRIPTION

The propose system is the Arduino based variable and compactable system with a single axis solar tracking system. The Arduino gets the data from the two LDR sensors to rotate the solar panel to the lighting place by using the servo motor. The voltage divider method is used as

the voltage sensor. The block diagram of single axis solar tracking system is shown in Figure 1.

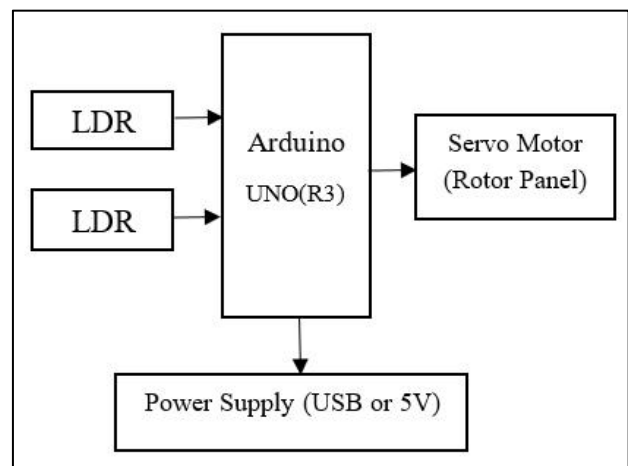


Figure 1. Block diagram of Solar Tracking System

2.1 Microcontroller

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.



Figure 2. Arduino Uno

2.2 Solar Panel

In this project, a small solar panel was used as a demonstrative element to visualize the tracking mechanism. The panel does not generate or store power, and was used purely as a showpiece to focus on the working of the tracking system through servo movement based on light intensity.



Figure 3. Solar panel

2.3 Servo Motor

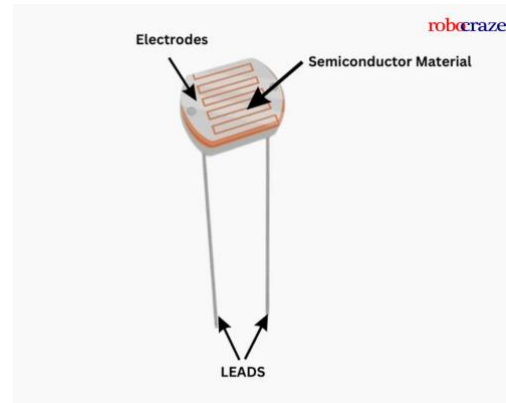
There are various types of motors such as AC, DC, Stepper, and Servo motors. In this project, a **servo motor** is used for its ability to provide precise angular movement. Servo motors are equipped with a control circuit that allows for accurate control of position, speed, and acceleration. They receive a control signal that determines the desired position of the shaft, and the internal circuitry ensures the motor moves to that exact angle. This makes servo motors ideal for applications like solar tracking, where controlled and smooth rotation is essential for aligning the solar panel with the sun's position.



Figure 4. Servo motor

1.1 LDR Sensors

The lighting sensor is composed of the fixed resistor and the light dependent resistor (LDR). Supply voltage is 5V, and the voltage of LDR cannot be connected directly to the controller. So, voltage divider method is applied to read the voltage of LDR. The value of voltage difference between two LDRs is compared to the sensitivity, because the motor cannot be rotated every time on the whole day. The solar tracking system is designed to rotate CW or CCW directions according to the greater level of voltage when the voltages difference of two LDRs. When the voltage difference is greater than or not the sensitivity, the motor rotates to its respective direction.



1.2 Overall Circuit Diagram

The overall circuit diagram is illustrated in Figure 7. There are three parts in the control systems; LDR sensor, microcontroller and servo motor. The LDR sense the sunlight and send the data to the microcontroller. The microcontroller control servo motor direction depend upon the signal on LDR sensors. The solar panel tracks the sunlight to have the maximum power and efficiency. The charge controller is used to control the voltage charging to the battery.

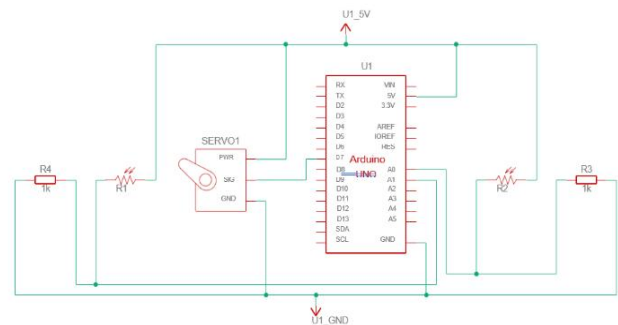


Figure 7. Overall Circuit of Single Axis Solar Tracking System

2. SOFTWARE IMPLEMENTATION

To simulate and compile the whole system, Arduino IDE software is established. Used Arduino IDE code is mentioned below.

```
#include <Servo.h>
#define servo_pin 7
int low_intensity = 100; // default threshold for LDR
int angle = 90; // starting angle of the servo
int targetAngle = 90; // target angle for the servo
int speed = 10; // lower value = slower servo speed (higher delay)
float LDR1, LDR2;
Servo Servo1;
void setup() {
    Serial.begin(9600);
    Servo1.attach(servo_pin);
    pinMode(A0, INPUT); // LDR1 sensor
    pinMode(A1, INPUT); // LDR2 sensor
    Servo1.write(angle); // initialize the servo position
    delay(5000); // wait for system to stabilize
}
void loop () {
```

```

LDR1 = analogRead(A0);
Serial.print("LDR1= ");
Serial.println(LDR1);
LDR2 = analogRead(A1);
Serial.print("LDR2= ");
Serial.println(LDR2);
if (LDR1>low_intensity && LDR2 >=
low_intensity) {
    delay(100); // Both LDRs are at high
intensity, centered position
    Serial.println("centered");
}
else if (LDR1 >= low_intensity && LDR2 <
low_intensity) {
    targetAngle = min(targetAngle + 1, 180);
// Limit target to 180 degrees
    moveToTargetAngle();
    Serial.println("angle++");
}
else if (LDR2 >= low_intensity && LDR1 <
low_intensity) {
    targetAngle = max (targetAngle - 1, 0);
// Limit target to 0 degrees
    moveToTargetAngle();
    Serial.println("angle--");
} else {
    delay (100);
}
}
void moveToTargetAngle() {
    if (angle < targetAngle) {
        angle++; // Move right (clockwise)
        Servo.write(angle); // Apply new angle
        delay(speed); // Delay to smooth out
movement
    } else if (angle > targetAngle) {
        angle--; // Move left (counter-
clockwise)
        Servo.write(angle); // Apply new angle
        delay(speed); // Delay to smooth out
movement
    }
}
}

```

3. TEST AND RESULT

Figure 10. shows that the entire project was designed and simulated using Tinkercad, an online platform for circuit design and Arduino simulation. The solar tracking system, including the Arduino Uno, LDR sensors, and servo motor, was successfully implemented and tested virtually. This allowed us to verify the functionality of the system, fine-tune sensor response, and visualize the panel's movement without physical hardware.

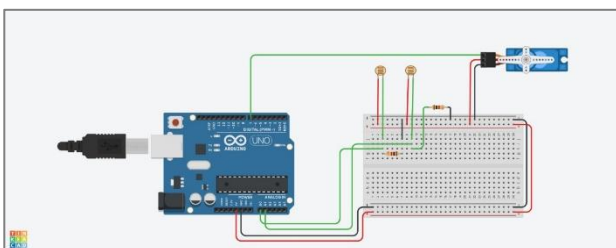


Figure 10. Software Implementation

Figure 11. shows implementation of the system is developed with solar panel.

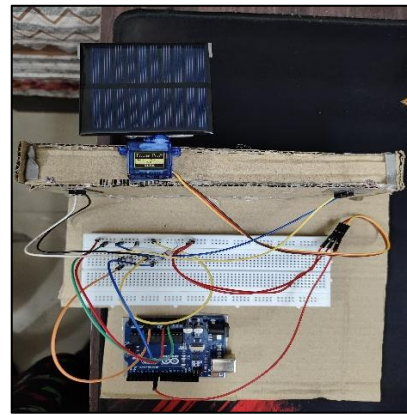


Figure 11. Testing of the whole system

4. CONCLUSION

In this project, a simple and low-cost single-axis solar tracking system was successfully designed and implemented using an Arduino UNO, two LDR sensors, and a servo motor. The system tracks the sun's position by comparing light intensity from both LDRs and adjusting the panel direction accordingly. This approach increases solar exposure throughout the day, potentially improving efficiency compared to fixed-position panels.

The simplicity, affordability, and reliability of this system make it ideal for small-scale applications and educational purposes, especially in rural or resource-limited areas. Although this version focuses on basic solar tracking, the concept can be further improved by adding dual-axis tracking, more efficient sensors, or integrating a power storage system. This project lays a strong foundation for future developments in renewable energy solutions, such as solar-powered devices and vehicles, contributing to a greener and more sustainable future.

5. ACKNOWLEDGMENTS

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6. REFERENCES

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